Workflow-Centric AI Governance

A Typed Gate Contract for Accountable Human-AI Decisions

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Abstract

Human oversight is frequently assigned as a role while the protocol that makes oversight meaningful remains underspecified. A person can be formally 'in the loop' while lacking admissible evidence, time, decisional authority, review capacity, or a usable fallback. This paper specifies a workflow-centric case contract for consequential human-AI decisions. Stage 1 maps three independent diagnostic states - each PASS, REVIEW, BLOCK, or UNKNOWN - plus distinct review-authority, review-capacity, fallback-authorization, and fallback-readiness inputs to one routing status. All-PASS yields EXECUTION ELIGIBLE only: execution is governed by a separate, versioned execution contract. Stage 2 records the actual current, in-scope signed disposition for cases routed to authorized review. A preliminary BLOCK is not a final denial; authority availability is not a decision event; UNKNOWN cannot silently become PASS; signed DEFER produces AUTHORIZED DEFER rather than reusing the Stage-1 EVIDENCE HOLD state; and fallback requires both authorization and operational readiness.

The v6.2 independent checker enumerates all 1,024 Stage-1 configurations and all 1,248 applicable Stage-2 configurations. The authored functions produce zero invariant violations, while nine isolated guard mutations each produce a counterexample, including a cross-stage mutation that collapses signed DEFER back into the Stage-1 EVIDENCE HOLD label. Exhaustive therefore means complete over the declared truth-table abstraction; it is not proof that the invariants are complete, comparison with external ground truth, or evidence of legal or field effectiveness. A shared fifteen-unit public-record corpus from official investigations of Robodebt, the Dutch childcare-allowance scandal, and Michigan unemployment-insurance claimant services is used only to show traceability of evidence, reasons, review access, authority, proportionality, feedback, and remedy.

The contribution is a falsifiable state-and-authority grammar for case routing. It does not establish that three gates are optimal, that the risk decomposition is exhaustive, that the routing priority is legally sufficient in any jurisdiction, or that the architecture improves outcomes. Its narrower claim is that missing evidence, unavailable capacity, nominal authority, and fallback readiness can be represented as different typed conditions rather than silently collapsed into approval or denial.

Keywords: AI governance; accountable decisions; contestability; human oversight; sociotechnical systems; typed protocols; workflow governance

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1. Introduction

AI governance often begins with model properties - data quality, accuracy, robustness, fairness, explainability, and monitoring. Those properties matter, but the model does not govern its own use. Consequential outcomes are produced by an institutional sequence that defines the task, selects evidence, configures interfaces, sets production expectations, allocates authority, routes exceptions, and turns a recommendation into action. A technically adequate model can therefore participate in an inadequately governed decision.

Formal human presence does not cure this defect. Automation can shift attention and response criteria in ways associated with complacency and automation bias (Parasuraman and Manzey, 2010). Cognitive forcing can reduce overreliance, but at measurable time and preference cost (Bucinca, Malaya, and Gajos, 2021). Effective oversight consequently depends on discriminability, incentives, task conditions, and an actual ability to intervene rather than the existence of a human job title (Langer, Baum, and Schlicker, 2025). Recent frameworks make the point still more directly: meaningful oversight requires defined roles, process architecture, epistemic capacity, cognitive space, decisional authority, and effective intervention channels (Gaube et al., 2026; van de Sande, Economou-Zavlanos, and van Genderen, 2026).

This paper shifts the primary unit of governance from the model alone to the decision workflow. The narrower research question is: can a common typed contract preserve distinct diagnostic functions while preventing preliminary machine states, missing evidence, or absent institutional capacity from silently becoming final action? The answer is a case-level routing specification. It deliberately does not solve the larger questions addressed by Parts II and III of the trilogy: whether a nominally compliant institution is losing governing capacity over time, and how a failing workflow can be contained and ended without abandoning remedy or decision responsibility.

The paper makes five bounded contributions. First, three diagnostic functions share one four-state type without being collapsed into one score. Second, a total triage function separates preliminary diagnostic states from routing status: all-PASS yields only EXECUTION ELIGIBLE, while a separate authorized-disposition function prevents authority availability from masquerading as a decision event in reviewed cases. Third, an append-only decision record connects each actual action to evidence, execution authority, fallback, and later correction. Fourth, both finite-stage semantics are exhaustively checked and mutation-tested within the declared finite abstraction. Fifth, the contract is placed against official public records and the current human-oversight literature to define both what is externally traceable and what remains unvalidated.

2. Relation to Existing Work and the Incremental Boundary

Evidence hierarchy. Peer-reviewed research and official investigations anchor established empirical and institutional claims. Recent 2026 preprints and working papers are used principally to map the current conceptual boundary and emerging terminology; they are not treated as independent validation of this contract, and claims that depend only on them remain provisional.

2.1 Sociotechnical accountability and workflow as unit of analysis

Internal algorithmic auditing has already extended accountability beyond model evaluation into organizational processes and documentation (Raji et al., 2020). Fairness research has shown that technical abstractions can erase social context and that invalid measurement can defeat an apparently rigorous analysis (Selbst et al., 2019; Jacobs and Wallach, 2021). These literatures establish that model performance cannot be separated from the institutional conditions under which a system is used.

The present contribution is narrower. It specifies what must remain explicit between a model output and a final institutional act: the state of required evidence, the nature of a diagnostic concern, the authority allowed to change an action, the availability of review capacity, the reality of fallback, and the disposition of cases that cannot be resolved on time.

2.2 Reliance, contestability, and practical authority

Human-AI performance is not captured by accuracy alone. Evaluation must distinguish appropriate reliance, complementary competence, safety, learning, and error recovery (Lai et al., 2023; Lee, 2026). Contestability likewise cannot be reduced to a post-hoc appeal form: meaningful intervention requires an opportunity, intelligible grounds, and an actor who can alter the decision (Almada, 2019). Elish's moral-crumple-zone analysis further warns that responsibility can be concentrated on a frontline human who lacks meaningful control (Elish, 2019).

The contract therefore separates diagnostic state from authority. A human reviewer who cannot slow, modify, reject, defer, or reach an independent authority does not supply effective arbitration merely by clicking an approval control.

2.3 Recent work narrows the novelty claim

By 2026, no defensible version of this paper can claim that it invented human gates, meaningful oversight, resource-constrained supervision, action-boundary control, or the requirement that humans retain understanding and intervention capacity. Gaube et al. (2026) provide a cross-disciplinary framework for effective human oversight; Langer, Lazar, and Baum (2026) explain why compliance testing must move beyond checklist evidence; Manheim and Homewood (2025) distinguish oversight from operational control and foreground feasibility limits; Azevedo (2026) formalizes uncertainty-aware oversight allocation; Lin et al. (2026) define a cognitive-integrity threshold below which oversight becomes merely procedural; and van de Sande et al. (2026) identify epistemic capacity, cognitive space, decisional authority, and intervention effectiveness as conditions for meaningful medical-AI oversight. Mazzocchetti (2026) separately specifies runtime action-boundary control with trusted provenance, fail-closed execution, and a settlement layer. These works remove any novelty claim based on 'humans must be able to intervene' or on separating a model proposal from executable action.

The boundary is narrower still after peer-reviewed 2026 work. Zhu et al. (2026) provide a layered-agency framework, a catalogue of oversight mechanisms, and end-to-end design patterns for meaningful human oversight. Calboreanu (2026) supplies a governance-first agent architecture with deterministic policy gates, execution authorization, escalation, and cryptographic audit. These contributions remove any remaining claim that structured rationales, circuit breakers, end-to-end oversight patterns, deterministic gates, or gated execution are novel here. The increment claimed in this paper is institutional and case-semantic: a cross-domain typed contract that preserves distinct diagnostic, routing, authority, capacity, fallback, and signed-disposition states around a consequential decision.

The incremental claim is narrower: a cross-domain, typed institutional case contract that distinguishes diagnostic state, routing status, authority credential, service capacity, fallback authorization, fallback readiness, and the actual signed disposition. Missing evidence remains UNKNOWN; preliminary machine states cannot directly become institutional denials; and unresolved cases remain visible rather than disappearing from performance denominators. The paper does not claim the trusted-provenance or execution-isolation contribution developed for agentic runtime systems. Whether this institutional contract improves real oversight remains an empirical question.

2.4 Regulatory and standards boundary

Article 14 of the EU AI Act requires effective human-oversight capabilities for applicable high-risk systems and addresses awareness of automation bias and the ability, where appropriate, to disregard, override, reverse, or interrupt outputs (European Union, 2024). NIST AI RMF 1.0 addresses roles, human-AI configurations, contextual limits, monitoring, appeal, override, incident response, recovery, and decommissioning (NIST, 2023). ISO/IEC FDIS 42105, at FDIS stage 50.00 in this revision, provides lifecycle guidance on human control and monitoring and explicitly extends ISO/IEC TS 8200 (ISO/IEC, 2026). This paper may supply evidence useful to selected practices in those instruments, but it does not establish legal conformity, standards conformity, regulatory sufficiency, or sector-specific compliance.

Established family	Already established	Additional specification here	Still unestablished
Effective human oversight	architectures, roles, processes, documentation	common four-state diagnostic type plus total case routing	comparative usability or effectiveness
Reliance / cognitive capacity	automation bias, calibration, cognitive forcing, comprehension conditions	UNKNOWN and capacity become explicit routing states	whether the contract preserves cognition
Oversight vs control	conceptual boundary and feasibility limits	preliminary diagnostic control separated from authorized disposition	security and organizational feasibility
Contestability / recourse	challenge, explanation, appeal, intervention	case record binds reasons, evidence, authority, fallback, and unresolved status	effect on real remedy
EU AI Act / NIST	oversight, override, appeal, lifecycle governance	auditable state and disposition evidence	legal sufficiency or conformity

Table 1. Incremental boundary relative to adjacent oversight and governance families.

3. Method and Claim Boundary

The paper follows a design-science logic: identify a recurrent governance failure, construct an artifact, expose its assumptions, and test whether the artifact behaves consistently under declared conditions (Hevner et al., 2004). The current revision deliberately removes the earlier simulation-heavy effectiveness rhetoric. The primary evaluation is now the finite contract itself; the external record exercise is used only for traceability of governance objects.

Evaluation	Question answered	Evidence produced	Expressly excluded
finite specification check	do both routing stages preserve stated invariants over their finite input spaces?	1,024 triage + 1,248 arbitration configurations; mutation counterexamples	completeness of the invariants; legal or outcome validity
public-record traceability	are evidence, reasons, review, authority, proportionality, feedback, and remedy visible in official records?	shared 15-unit source-bounded corpus	case classification, causation, prevalence, prevention
literature-boundary audit	does the claimed increment survive direct comparison with current adjacent work?	explicit already-established / added / unestablished table	novelty validation by citation count or priority claim

Table 2. Evaluation questions, evidence supplied, and expressly excluded claims.

Four claims are advanced and no stronger ones: (1) architectural - distinct diagnostics can share one state type without erasing their different meanings; (2) internal safety - the authored triage and arbitration functions enforce their stated invariants within the finite abstractions; (3) traceability - the workflow objects appear in independent official records; and (4) design discipline - unresolved evidence, absent authority, unavailable capacity, invalid signatures, and imaginary fallback can be represented as explicit governance states rather than silently normalized.

4. Domain Contract

Before any gate is used, the institution defines a versioned domain contract. The contract states the protected decision function, admissible inputs, model recommendations, permissible institutional actions, hard constraints, value and impact questions, temporal-validity rules, role and authority matrix, fallback process, review service levels, record-retention rules, and monitoring plan. A recommendation is not an institutional action; the contract must state the transition between the two.

Contract object	Minimum content	Failure condition
decision scope	who is affected; consequences; excluded uses	scope drifts without review
evidence contract	required fields, provenance, missingness, time validity	required evidence missing or uninterpretable
authority matrix	who may execute, modify, stop, defer, escalate, or override	reviewer or executor is nominally responsible but cannot act
execution contract	who may act after all-PASS; scope, freshness, revocation, separation of duties, witness/log requirements	EXECUTION ELIGIBLE is mistaken for execution or approval
capacity contract	review queue, protected time, service deadline	routed work exceeds available review capacity
fallback	approved non-model or alternate service path; readiness test	fallback exists on paper but cannot carry the workload
record and remedy	reason, evidence, notice, appeal, correction, retention	decision cannot later be reconstructed or challenged

Table 3. Minimum objects in the versioned domain contract and corresponding failure conditions.

The domain contract must state whether an all-PASS proposal may be executed automatically, requires an additional human or service authorization, or must enter another domain-specific control. It must also specify the executor, permissible action, scope, credential freshness and revocation, separation-of-duties rules, audit witness or logging, and failure behavior. EXECUTION ELIGIBLE is therefore never self-executing. The architecture is inappropriate where the institution cannot define a lawful fallback, cannot grant reviewers or executors effective authority, cannot retain the minimum evidence lawfully, or cannot resource review without systematic pressure to bypass it. In those conditions the governance answer may be non-deployment, narrowing, or suspension rather than interface refinement.

The case-time inputs r, c, b, and f are route-local availability variables; they are not a substitute for deployment-level readiness. Before the case contract is activated, the domain contract must separately state which capabilities - including fallback, review service, identity/authorization services, record infrastructure, and emergency handling - are mandatory as standing deployment preconditions. A domain may therefore prohibit execution even when Stage 1 returns EXECUTION ELIGIBLE if a required deployment-level readiness condition is not met. This separation prevents a truth-table flag from being mistaken for a complete operating-readiness judgment.

5. Three Diagnostic Gates and Authorized Arbitration

Every upstream diagnostic returns one state from S = {PASS, REVIEW, BLOCK, UNKNOWN}. PASS authorizes progression to the next diagnostic, not final action. REVIEW identifies a question requiring authorized judgment. BLOCK pauses the proposed action and routes the case to arbitration; it is preliminary, not a final denial. UNKNOWN means that required evidence is unavailable, stale, contradictory, or uninterpretable under the domain contract; it routes evidence completion or fallback rather than being imputed as PASS.

5.1 Gate 1 - hard constraints and admissibility

Gate 1 tests non-negotiable constraints: legal prohibitions, scope restrictions, minimum evidentiary preconditions, and other hard rules specified by the domain. It is not a model-risk score. A BLOCK means that the proposed action cannot progress without authorized disposition; UNKNOWN means that the predicate itself cannot be evaluated with the available evidence.

5.2 Gate 2 - value, impact, and conflict

Gate 2 surfaces material value tension, proportionality concerns, conflicting objectives, and distributional or rights-sensitive issues that should not be collapsed into a single optimization score. It can return REVIEW when tradeoffs require contextual judgment even if no hard prohibition is present. A domain that cannot state which value questions matter should not hide that absence behind a generic ethics score.

5.3 Gate 3 - temporal validity and evidence reliability

Gate 3 asks whether the evidence, policy, system version, and relevant assumptions are current enough for the decision. Time validity is a governance property because a formally complete record can still be stale. The gate distinguishes 'known adverse' from 'unknown because the basis is no longer trustworthy.'

5.4 Authorized human arbitration

A non-PASS diagnostic first creates a routing requirement, not a decision. Where evidence is missing and an authorized, resourced review function exists, the case enters EVIDENCE HOLD for evidence completion and rerun. Where REVIEW or BLOCK remains without UNKNOWN and an authorized, resourced review function exists, the case enters AUTHORIZED REVIEW. The arbitrator receives the recommendation, gate reasons, relevant evidence, uncertainty, permissible actions, fallback status, and the scope of their authority. 'Reviewed' is not a valid terminal outcome.

Only Stage 2 can convert AUTHORIZED REVIEW into an institutional disposition. Its required event is an explicit instruction q in {RELEASE, MODIFY, REJECT, DEFER, INVOKE FALLBACK}, signed or otherwise authenticated by a named decision authority, current at execution, and within that actor's scope. The record includes the instruction, rationale, evidence considered, unresolved uncertainty, timestamp, credential and scope, and accountable role. Availability of an authority is not a proxy for that event. A preliminary BLOCK can therefore be released, modified, rejected, or deferred after judgment without any contradiction: only a valid signed REJECT yields FINAL STOP.

The Boolean validity input v is an interface boundary, not a claim that authentication is solved. The domain contract must define accepted signature or authentication methods, credential issuance and revocation, scope, freshness, dual-control requirements where applicable, and behavior when the identity or execution service is unavailable. The finite checker assumes that upstream validation is truthful; it does not test identity proofing, cryptography, coercion, or the legal validity of electronic signatures.

Review authority and review capacity are separate inputs; fallback authorization and fallback operational readiness are also separate. If either member of a pair is absent, that route is unavailable. Capacity failure is an institutional outcome, not missing data.

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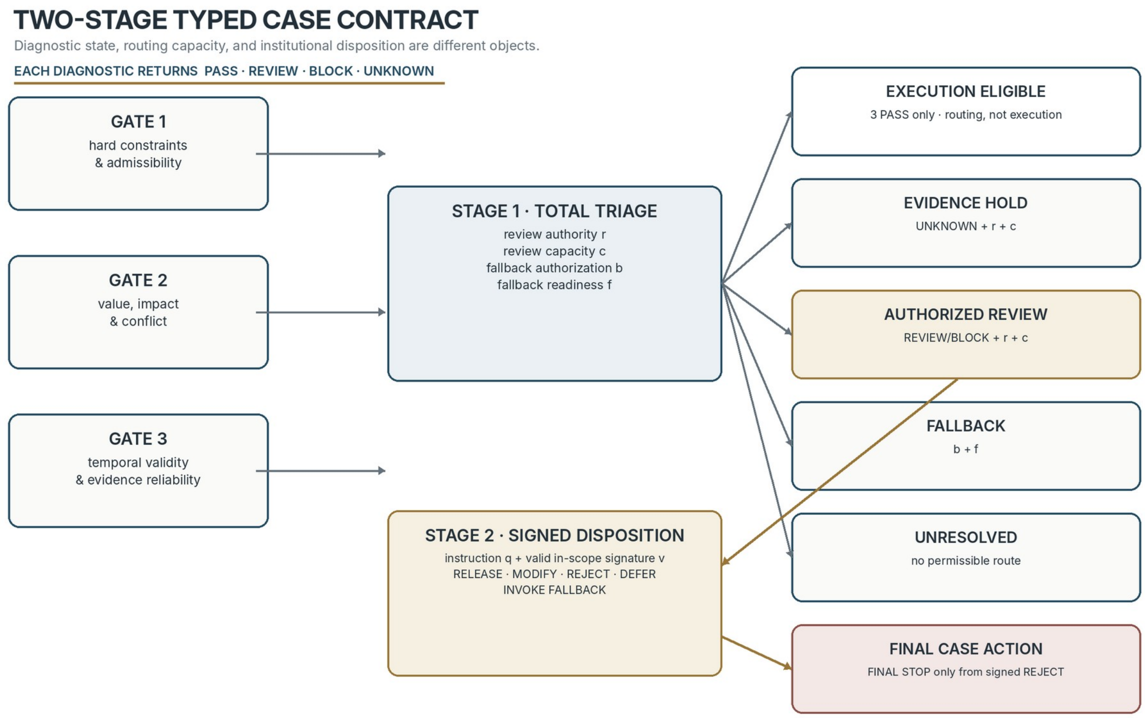

Figure 1. Stage 1 produces routing status only. All-PASS means EXECUTION ELIGIBLE, not execution; reviewed cases require a valid signed Stage-2 disposition, and every actual action must be separately authorized and recorded.

6. Total Routing Semantics and Finite-Domain Verification

6.1 Stage 1 - total triage function

Let s = (S1, S2, S3) be the three diagnostic states. Let r indicate a valid review-authority credential for the case, c available review capacity, b valid authorization to invoke fallback, and f verified fallback readiness. The total triage function T(s,r,c,b,f) returns one routing status from {EXECUTION ELIGIBLE, EVIDENCE HOLD, AUTHORIZED REVIEW, FALLBACK, UNRESOLVED}. EXECUTION ELIGIBLE is a typed permission to enter the domain-defined execution control; it is not the action itself.

1. PASS/PASS/PASS produces EXECUTION ELIGIBLE. No institutional flag can convert a non-PASS diagnostic tuple into execution eligibility, and the status cannot by itself execute or approve the proposal.
2. If any gate is UNKNOWN and r = c = 1, route to EVIDENCE HOLD for evidence completion and rerun. Missing required evidence cannot be inferred away.
3. Otherwise, if at least one REVIEW or BLOCK remains and r = c = 1, route to AUTHORIZED REVIEW. The status records that judgment is pending; it is not a final action.
4. If the required evidence/review route is unavailable and b = f = 1, route to FALLBACK.
5. Otherwise return UNRESOLVED.

The priority is an authored design choice, not a universal legal rule. A domain that requires immediate containment or a different relation between a hard prohibition and missing evidence must encode and verify that alternative. Stage 1 has four invariants: EXECUTION ELIGIBLE requires three PASS states; EVIDENCE HOLD requires UNKNOWN plus review authority and capacity; AUTHORIZED REVIEW requires a non-UNKNOWN REVIEW/BLOCK trigger plus review authority and capacity; and FALLBACK requires both fallback authorization and readiness. A fifth cross-boundary requirement is non-formal: no Stage-1 status is itself an execution event.

6.2 Stage 2 - signed institutional disposition

Stage 2 applies only after an eligible non-UNKNOWN diagnostic pattern has reached AUTHORIZED REVIEW. Let q be the recorded instruction in {NONE, RELEASE, MODIFY, REJECT, DEFER, INVOKE FALLBACK}; let v indicate that the instruction is signed, current, and within authority at execution. The disposition function A(q,v,b,f) returns AUTHORIZED REVIEW while q = NONE or v = 0. With v = 1 it maps RELEASE to RELEASE, MODIFY to MODIFIED ACTION, REJECT to FINAL STOP, DEFER to AUTHORIZED DEFER, and INVOKE FALLBACK to FALLBACK only when b = f = 1, otherwise UNRESOLVED. AUTHORIZED DEFER is a signed institutional disposition that holds the proposed action pending a specified condition, evidence request, or future review; it is intentionally distinct from Stage-1 EVIDENCE HOLD, which is triggered by UNKNOWN evidence state.

Stage 2 has four central invariants: every action-changing disposition requires a valid signature; FINAL STOP requires the actual signed instruction REJECT, not merely a BLOCK diagnostic or available authority; AUTHORIZED DEFER requires the actual signed instruction DEFER and cannot be emitted by Stage 1; and FALLBACK requires a signed invocation plus fallback authorization and readiness. Conversely, Stage-1 EVIDENCE HOLD cannot be emitted by Stage 2. This is the paper's critical separation between evidence state, competence, availability, and decision.

6.3 Worked end-to-end example

Consider a synthetic public-benefit eligibility review; it is an illustration of the contract, not a reconstruction of Robodebt, the Dutch inquiry, Michigan claimant services, or any legal rule. A model proposes suspension after detecting an earnings discrepancy. Gate 1 finds the case within the program and the proposed action type admissible (PASS). Gate 2 finds a material proportionality question because a verified household circumstance may change the permissible response (REVIEW). Gate 3 finds that the earnings record is older than the domain contract allows (UNKNOWN). The institution has a competent reviewer and available review capacity, and its non-model fallback is both authorized and ready: (r,c,b,f) = (1,1,1,1).

Step	Declared inputs	Contract output	Why
initial Stage 1	s = (PASS, REVIEW, UNKNOWN); r=c=b=f=1	EVIDENCE HOLD	required evidence is unresolved; UNKNOWN has priority over merits review
evidence completion	current earnings record obtained; Gate 3 rerun to PASS	rerun Stage 1	the missing predicate is evaluated rather than imputed
second Stage 1	s = (PASS, REVIEW, PASS); r=c=1	AUTHORIZED REVIEW	a value-sensitive question remains and a resourced authority exists
Stage 2	q = MODIFY; v=1; scope and freshness valid	MODIFIED ACTION	the signed instruction, not the preliminary REVIEW, creates the action
counterfactual guard	q = REJECT; v=0	AUTHORIZED REVIEW	an unsigned rejection cannot become FINAL STOP

Table 4. Worked end-to-end routing example; the case is synthetic and does not reconstruct an investigated system.

The UNKNOWN-first priority protects against deciding a merits question while mandatory evidence is stale or missing. That rationale is defeasible. In time-critical domains, a domain contract may require an immediate reversible safe hold, emergency treatment, or another priority; the alternative must be stated as a different total function and verified under its own invariants rather than silently overriding this one.

6.4 Exhaustive results and analytic cross-check

The executable checker enumerates 4^3 × 2^4 = 1,024 Stage-1 configurations and 26 eligible non-UNKNOWN diagnostic patterns × 6 instructions × 2^3 execution flags = 1,248 Stage-2 configurations. Both unmutated functions produced zero authored-invariant violations. Counts are properties of uniform logical enumeration, not estimates of real-world frequency.

The Stage-1 totals can also be derived without running code. Of 4^3 = 64 diagnostic tuples, 3^3 = 27 contain no UNKNOWN, so 37 contain at least one UNKNOWN; subtracting the all-PASS tuple leaves 26 non-UNKNOWN tuples requiring judgment. The all-PASS tuple varies over all 16 institutional-flag combinations, yielding 16 EXECUTION ELIGIBLE states. Setting r=c=1 leaves four b/f combinations, yielding 37×4=148 EVIDENCE HOLD and 26×4=104 AUTHORIZED REVIEW states. Among the remaining twelve r/c/b/f combinations, three have b=f=1 and nine do not, yielding 63×3=189 FALLBACK and 63×9=567 UNRESOLVED states.

For Stage 2, each of the 26 eligible tuples has 6×2^3=48 instruction/flag combinations. AUTHORIZED REVIEW covers q=NONE for all eight flag combinations plus each of the five non-NONE instructions when v=0 for four b/f combinations: 26×(8+20)=728. RELEASE, MODIFY, REJECT, and DEFER each occupy 26×4=104 valid-signature combinations, with DEFER reported as AUTHORIZED DEFER. A valid fallback instruction succeeds only at b=f=1, giving 26 FALLBACK and 26×3=78 UNRESOLVED states. This independent combinatorial derivation reproduces the executable counts exactly.

Stage / status	Count	Interpretation
Stage 1 · EXECUTION ELIGIBLE	16	all three diagnostics PASS; routing only; four institutional flags vary
Stage 1 · EVIDENCE HOLD	148	UNKNOWN with review authority and capacity
Stage 1 · AUTHORIZED REVIEW	104	REVIEW/BLOCK without UNKNOWN, with authority and capacity
Stage 1 · FALLBACK	189	required primary route unavailable; fallback both authorized and ready
Stage 1 · UNRESOLVED	567	no permissible operational route under declared inputs
Stage 2 · AUTHORIZED REVIEW	728	no instruction or invalid/out-of-scope signature; judgment remains pending
Stage 2 · each direct result	104	RELEASE, MODIFIED ACTION, FINAL STOP, and AUTHORIZED DEFER each
Stage 2 · FALLBACK / UNRESOLVED	26 / 78	signed invocation succeeds only with authorization and readiness

Table 5. Exhaustive finite-domain output counts under uniform logical enumeration.

6.5 Mutation counterexamples

Weakened guard	Minimal configuration	Unsafe result	Violated invariant
UNKNOWN treated as PASS	PASS / PASS / UNKNOWN; all institutional flags absent	EXECUTION ELIGIBLE	execution eligibility requires 3 PASS
review without authority	PASS / PASS / REVIEW; capacity present; authority absent	AUTHORIZED REVIEW	review requires authority + capacity
review without capacity	PASS / PASS / REVIEW; authority present; capacity absent	AUTHORIZED REVIEW	review requires authority + capacity
fallback without authorization	PASS / PASS / REVIEW; fallback ready only	FALLBACK	fallback requires authorization + readiness
fallback without readiness	PASS / PASS / REVIEW; fallback authorized only	FALLBACK	fallback requires authorization + readiness
unsigned rejection	q = REJECT; signature invalid	FINAL STOP	final stop requires signed rejection
fallback disposition without authorization	signed invocation; ready but unauthorized	FALLBACK	signed invocation + authorization + readiness
fallback disposition without readiness	signed invocation; authorized but unready	FALLBACK	signed invocation + authorization + readiness
Stage-2 defer-state collision	q = DEFER; signature valid	EVIDENCE HOLD	Stage 2 must emit AUTHORIZED DEFER, not Stage-1 EVIDENCE HOLD

Table 6. Isolated guard mutations and minimal counterexamples, including the v6.2 cross-stage defer-state test.

The counterexamples establish necessity only relative to the authored invariants. They do not establish completeness of the four-state type, optimality of the priority order, validity of gate inputs, truthfulness of credentials or organizational records, accessibility of review, or legal sufficiency. Those are external questions.

7. Decision Record and Accountability

Every consequential case should generate an append-only decision record sufficient to reconstruct the transition from recommendation to institutional action. At minimum it contains contract and model versions; recommendation and proposed action; gate inputs, outputs, and reason codes; Stage-1 routing status; evidence provenance and temporal status; review, execution, and fallback credentials and their scopes; the actual execution event or Stage-2 instruction and its authority validity at execution; fallback or override; decision latency; notice and challenge route; downstream correction where lawfully measurable; and any later amendment. For an all-PASS case, the record must distinguish the EXECUTION ELIGIBLE timestamp from the later action timestamp and identify the rule and actor or service that authorized execution.

The record is not a demand to retain every datum indefinitely. Retention, access, minimization, security, and deletion are domain-specific controls. A useful implementation can separate operational, audit, and notice views while preserving a verifiable chain of responsibility.

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Role	Non-delegable accountability	Illustrative evidence
Policy owner	purpose, legal basis, protected interests, exception policy	approved domain contract and change record
System owner	model limits, versioning, technical monitoring	validation, incident, and release records
Workflow owner	staffing, queue, thresholds, fallback, bypass controls	workload and routing review
Case reviewer	case decision within actual authority	case-specific decision record
Independent / competent authority	escalation, suspension or other action within mandate	signed determination and reviewable rationale

Table 7. Non-delegable accountability roles and illustrative evidence.

This allocation does not eliminate shared responsibility. Its purpose is to prevent leadership decisions about capacity, interface design, thresholds, or fallback from being reframed as the fault of the last human operator.

8. Public-Record Traceability Illustrations

The trilogy uses a shared, frozen corpus of fifteen passage-level evidence units from three official investigations: Australia's Robodebt Royal Commission, the Dutch parliamentary inquiry into the childcare-allowance scandal, and Michigan's unemployment-insurance claimant-services audit (Royal Commission into the Robodebt Scheme, 2023; Childcare Allowance Parliamentary Inquiry Committee, 2020; Michigan Office of the Auditor General, 2016). The cases are purposively selected negative illustrations. They are not a representative sample, do not classify cases under this contract, and cannot establish that the architecture would have prevented harm.

Contract object	Independent official trace	Design implication	Unsupported inference
sufficient evidence	Michigan audit documents fact-gathering and reason-giving deficiencies in sampled determinations	UNKNOWN must remain actionable; provenance and missingness belong in the record	that a typed gate would have produced the correct decision
facts, reasons, review	Michigan and Robodebt records emphasize reasons and usable review pathways	challenge requires intelligible grounds, not merely a nominal appeal channel	comparative effect on due process or appeal success
lawful authority and records	Robodebt recommendations emphasize legal foundations, documented assumptions, scrutiny, and important decision records	thresholds, routing status, credentials, and signed dispositions need versioned ownership	that documentation alone creates lawful oversight
proportionality / individual circumstances	Dutch inquiry documents rigid group-based and all-or-nothing administration	categorical signals cannot by themselves settle value-sensitive action	that one generic value gate captures the legal standard
remedy and continuity	Robodebt and Dutch records foreground dispute handling and delayed compensation	unresolved cases, delays, and remedy remain governance outcomes	that the routing order is sufficient for remedy in any jurisdiction

Table 8. Public-record traceability illustrations and unsupported inferences.

The evidence supports traceability only: independent official investigations treat evidence, reasons, review, authority, proportionality, records, and remedy as consequential workflow objects. The same records also show the limit of the artifact. Typed states cannot decide whether a legal basis is valid, a reason is adequate, a review pathway is accessible, or a protected interest has been weighed legitimately.

The same frozen fifteen-unit corpus is reused in Parts II and III to keep source locators and coding lineage comparable across the trilogy. That economy creates a theory-first narrative-fit risk: the same negative records cannot serve as three independent validations merely because each paper asks a different traceability question. No cross-paper convergence, triangulation, replication, or cumulative validation is claimed. Each theory requires new positive, negative, and boundary cases selected independently of its own terminology.

9. Deployment Evaluation and Failure Criteria

A field implementation should be evaluated as a sociotechnical intervention, not by asking whether the software obeys the routing table. Before operational use, the institution should pre-specify at least the following tests:

1. construct validity for each diagnostic input and reason code, including subgroup and accessibility effects;
2. reviewer comprehension of PASS, REVIEW, BLOCK, and UNKNOWN and the distinction between preliminary and final action;
3. authority validity: whether named actors can actually obtain records, modify action, pause a queue, and reach an independent escalation route;
4. capacity adequacy under realistic demand, including queue delay and protected review time;
5. fallback capacity and service-continuity exercises under partial and full interruption;
6. record completeness, contestability, privacy, and lawful retention;
7. comparative burden against a simpler workflow and explicit evidence that added friction improves a declared protected function; and
8. sunset conditions when the contract becomes ritual, burdensome, or inferior to a lower-complexity alternative.

The architecture should be rejected, narrowed, or suspended if it creates systematic unresolved queues; if reviewers become responsible without real authority; if UNKNOWN is routinely overridden as an administrative convenience; if fallback is nominal; if reasons become boilerplate; or if a simpler process performs at least as well with lower burden. Structured friction is defensible only when it protects a declared portfolio of outcomes and remains open to independent challenge.

10. Relationship to Parts II and III

The trilogy uses different levels of analysis and they must not be collapsed. Part I is case-level: it asks how a proposed action is routed at the point of decision. Part II is longitudinal and institutional: it asks whether visible procedural fidelity can coexist with a persistent loss of independent judgment, governing capacity, and a protected function. Part III is transition-level:

it asks how signals are investigated, how exposure may be contained, and how a workflow is retired and decision capacity is reconstituted or responsibly disposed.

Part I term	Meaning	Not equivalent to
BLOCK	preliminary diagnostic state on a proposed case action	final denial; incident; retirement
FINAL STOP	actual signed REJECT disposition of the proposed case action	BLOCK; available authority; Proper Ending of the workflow
UNRESOLVED	case cannot be completed under current evidence/capacity/fallback state	proof of PAC
persistent routing failures	possible governance trace requiring investigation	automatic GDI candidate or retirement authority

Table 9. Part I states and their non-equivalence to Part II/III institutional classifications.

No Part I state automatically establishes Pre-Abuse Collapse, triggers a Circuit Breaker, or authorizes Proper Ending. Those determinations require the evidentiary and authority boundaries specified in Parts II and III (Sato, 2026a, 2026b).

11. Limitations

The model is a finite specification, not a representation of a full organization. Its inputs are assumed to be correctly classified; it does not verify the semantic adequacy of evidence, legal rules, value judgments, or authority. The same organization that misreports capacity or fallback readiness can make a formally safe routing function operationally unsafe.

The three-diagnostic decomposition and two-stage priority are deliberately not claimed as universal or optimal. Some domains require additional diagnostics, emergency containment, multiple signatures, appeal before execution, or a different priority order. Others may be safer with fewer state transitions. Any extension should preserve the distinction among preliminary diagnosis, routing status, credentialed authority, and an actual disposition event and should be subjected to its own exhaustive or property-based tests.

The public-record corpus is narrow, negative, purposive, and author-coded. It supports source traceability but not causal identification, prevalence, specificity, or comparative effectiveness. No participant, field, or prospective operational data are analyzed.

The contract can itself create harm through delay, surveillance, bureaucratic ritual, or over-documentation. A workflow that prevents one class of error by making lawful service inaccessible is not successful. Governance therefore requires service, remedy, and capacity outcomes in addition to formal state correctness.

12. Conclusion

A human decision stage is not meaningful oversight merely because a person appears at the end of an AI-assisted workflow. The institution must preserve evidence, distinguish diagnostic uncertainty from substantive conflict, allocate real decisional authority, resource the review it demands, provide an operational fallback, and keep unresolved cases visible.

The revised two-stage contract makes those conditions testable. All 1,024 triage configurations and all 1,248 applicable arbitration configurations satisfy the authored safety invariants, while each of nine isolated guard mutations yields a counterexample. In particular, all-PASS produces only EXECUTION ELIGIBLE; no route label is treated as an action. FINAL STOP likewise cannot be inferred from a diagnostic BLOCK plus the mere availability of authority; it requires an actual valid signed rejection. The result is intentionally narrow: it establishes internal contract consistency, not field effectiveness or legal sufficiency.

The practical contribution is therefore a state-and-authority grammar for accountable case routing. It defines the front door of the trilogy. Part II examines how an institution can preserve that visible structure while its governing capacity erodes; Part III specifies how a suspected failure is investigated, contained, ended, and handed back without converting an indicator into the decision maker.

Appendix A. Trilogy Terminology Map

The same terms are used at different levels of analysis across the trilogy. This map prevents a diagnostic signal or preliminary state from acquiring decision authority by terminology alone.

Term	Primary level	Meaning and authority boundary
Gate state	case / Part I	PASS, REVIEW, BLOCK, or UNKNOWN; a diagnostic output, never a final institutional act
Routing status	case / Part I	Stage-1 destination such as EXECUTION ELIGIBLE, EVIDENCE HOLD, AUTHORIZED REVIEW, FALLBACK, or UNRESOLVED; never an act by itself
Execution event	case / Part I	separately authorized downstream act after EXECUTION ELIGIBLE; scope, freshness, revocation, and audit rules come from the domain contract
Signed disposition	case / Part I	actual current, in-scope Stage-2 instruction; only signed REJECT can create FINAL STOP
Authorized defer	case / Part I	signed Stage-2 DEFER disposition; distinct from UNKNOWN-origin EVIDENCE HOLD and never a final denial
Incipient capacity-loss risk	institution / Part II	credible judgment/capacity decline without the full realized-state evidence
PMGCL	institution / Part II	descriptive four-part state: stable visible procedure with persistent judgment loss, governing-capacity loss, and protected-function impairment
PAC subtype	institution / Part II	provisional etiological subtype: PMGCL plus an identified materially adverse ordinary-exposure contrast under a predeclared malicious-pathway intervention/reference condition; not proof that malice is absent or causally unnecessary
GDI	monitoring / Part III	defeasible governance trace that may route investigation but cannot authorize action
Circuit Breaker	transition / Part III	authorized, scoped, reversible containment; not retirement or closure
Resolution Collapse	transition / Part III	authorized finding that approved remediation cannot restore operation within the declared horizon
Proper Ending	institution / Part III	evidence- and remedy-preserving retirement process; not merely switching off a model

Term	Primary level	Meaning and authority boundary
Authority Return	institution / Part III	reconstitution, reassignment, or residual custody of decision capacity and unresolved obligations
Validity hold	measurement / Part III	preserves a signal for investigation while blocking automated escalation until the measurement basis is repaired

Table A1. Trilogy terminology map and authority boundaries.

Data, Code, and Reproducibility

Version 6.2 includes an independent reimplementation, verify_part1_contract_v62.py, derived from the published finite specification rather than by editing the historical v5.9 checker. It enumerates 1,024 Stage-1 configurations and 1,248 applicable Stage-2 configurations. The v6.2 semantics reproduce the Stage-1 distribution 16 / 148 / 104 / 189 / 567 and the Stage-2 distribution 728 / 104 / 104 / 104 / 104 / 26 / 78 across AUTHORIZED REVIEW, RELEASE, MODIFIED ACTION, FINAL STOP, AUTHORIZED DEFER, FALLBACK, and UNRESOLVED respectively, with zero authored-invariant violations. Nine isolated mutations - the historical eight guard weakenings plus a cross-stage DEFER-to-EVIDENCE-HOLD collision - each produce a counterexample. No field, participant, employer, client, or confidential data are introduced. The shared public-record evidence matrix remains a traceability artifact rather than validation data.

Generative AI Use Disclosure

OpenAI ChatGPT/Codex assisted with literature discovery, adversarial review, structural and language revision, independent finite-specification reimplementation, document generation, and quality assurance. An Anthropic Claude critique supplied by the author was used as an additional adversarial-review input; its combinatorial, bibliographic, and substantive claims were independently checked before adoption. Version 6.2 adds a cross-stage semantic invariant, a ninth mutation test, an explicit deployment-readiness boundary, current oversight-standard context, and publication QA. The finite checks remain internal specification tests rather than field validation. No participant or field data were supplied. The author retains intellectual and editorial control and is responsible for the claims, citations, analyses, and conclusions in the public version.

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